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Analysis of Scapular Kinematics and Muscle Activity by Use of Fine-Wire Electrodes During Shoulder Exercises

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Background: During nonoperative or postoperative rehabilitation after sports injuries, exercise selection is often based on minimal load on the injured/repaired glenohumeral structures, while optimally activating scapulothoracic muscles. Previous research explored scapular muscle activity during rehabilitation exercises using surface electromyography (EMG). However, limited information exists about the deeper lying muscle activity, measured with fine-wire electrodes, even more in combination with 3-dimensional scapular kinematics.

Purpose: To report scapular kinematics synchronously with surface and fine-wire EMG during specific shoulder exercises for early rehabilitation.

Study Design: Descriptive laboratory study.

Methods: A total of 23 healthy male patients were recruited. Three-dimensional scapular kinematics were measured combined with EMG recording of 8 muscles during 4 commonly used shoulder exercises (inferior glide, low row, lawnmower, and robbery). Upper (UT), middle (MT), and lower (LT) trapezius and serratus anterior muscle activities were measured with bipolar surface electrodes. Intramuscular electrodes were placed in the levator scapulae (LS), rhomboid major (RM), pectoralis minor (Pm), and infraspinatus (IS) muscles. All data were normalized as a percentage of maximal voluntary isometric contraction (%MVIC). A linear mixed model with Bonferroni correction was applied for statistical analysis.

Results: Scapular kinematics revealed an anterior tilt position during the inferior glide, low row, and robbery ($P < .05$). An upward rotation position between 20° and 30° was reached in all exercises except low row. Inferior glide (31°) and low row (42°) represented a significantly increased internal rotation position compared with lawnmower and robbery. Lawnmower and robbery showed significantly ($P < .05$) more MT (lawnmower, 36% MVIC; robbery, 39% MVIC) and RM (lawnmower, 59% MVIC; robbery, 66% MVIC) activation compared with inferior glide and low row. Lawnmower and robbery showed significantly ($P < .05$) less Pm activation (9.5%-12% MVIC). LS was significantly more active during robbery (58% MVIC) compared with inferior glide and low row (27%-36% MVIC) ($P < .05$). IS showed moderate activity (24%-37% MVIC) for all exercises, except low row (13% MVIC).

Conclusion/Clinical Relevance: This study provides new insights about scapular positions and activation of the deeper layer muscles during 4 commonly used shoulder rehabilitation exercises. The lawnmower showed a favorable position of the scapula with less Pm activity in contrast to the low row. The inferior glide, lawnmower, and robbery should not be implemented in early phases of shoulder rehabilitation because of their moderate muscle activity.

Keywords: shoulder; physical therapy/rehabilitation; electromyography; rotator cuff; scapular kinematics

During the early phases of rehabilitation after a sports injury, or the protective rehabilitation phase after shoulder surgery, exercise selection is often based on minimal load on the injured/repaired glenohumeral structures, with optimal activation of the scapulothoracic muscles. As an example, high activity of the healed rotator cuff (RC) muscles is not tolerated in the early rehabilitation phase after RC repair. Alternatively, exercise therapy is initiated with activation of the scapular muscles, as scapular dyskinesis appears to be a nonspecific response to shoulder dysfunction.[9] Therefore, early rehabilitation exercises often focus on strengthening the scapular muscles during isometric exercises or in small ranges of motion.[10]

Many factors determine the selection of exercises, such as the patient's goals and expectations, the proven effectiveness of a specific program, the tissue irritability, or the electromyography (EMG) activity in specific muscles. EMG has been widely used to examine muscle activity during commonly performed rehabilitation exercises. The systematic review by Schory et al[16] made an overview of 15 observational studies exploring the muscle activity in the scapular muscles during exercises, all using surface EMG. They proposed a selection of exercises to target

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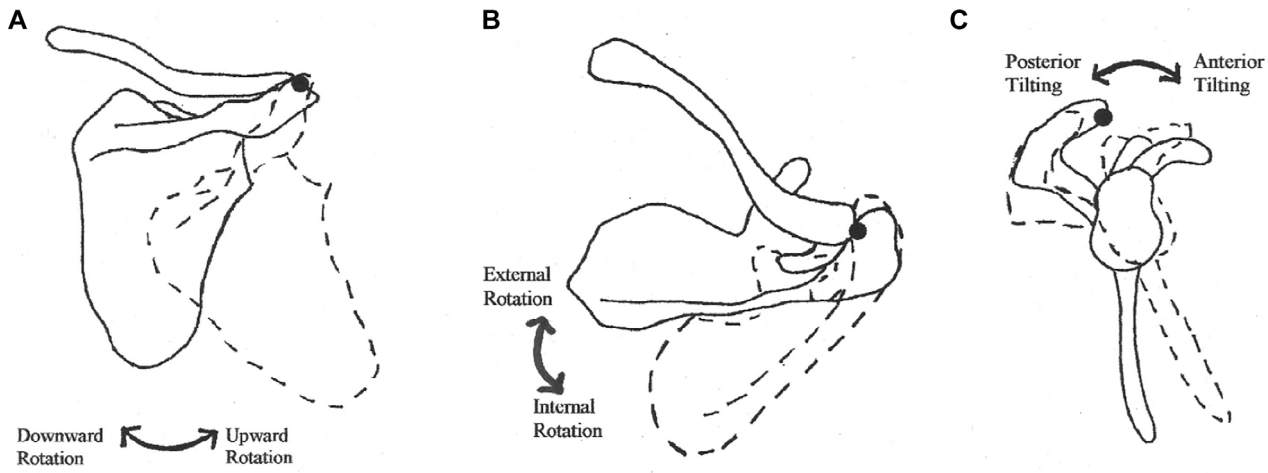

Figure 1. Scapular rotations: (A) downward/upward rotation around the frontal axis (*Y*-axis); (B) internal/external rotation around the vertical axis (*Z*-axis); and (C) anterior/posterior tilt around the horizontal axis (*X*-axis). Modified and reprinted from Ludewig PM, Hoff MS, Osowski EE, Meschke SA, Rundquist PJ. Relative balance of serratus anterior and upper trapezius muscle activity during push-up exercises. *Am J Sports Med*. 2004;32(2):484-493.[12]

activity in a specific muscle. However, surface EMG is limited in use since only superficial muscles can be reached. In contrast, deeper layer muscles may be explored by using fine-wire EMG. Castelein et al[5,6] used fine-wire electrodes combined with surface EMG to examine the muscle recruitment during shoulder exercises, offering new insights into muscle recruitment of the pectoralis minor (Pm), levator scapulae (LS), and rhomboid major (RM) muscles. However, those studies are limited to a number of exercises without exploring the 3-dimensional (3D) kinematics of the scapula.

The knowledge about 3D scapular behavior with respect to the thorax during shoulder exercises provides an additional value in the exercise's choice within a rehabilitation program.[19,20] The scapula rotates around 3 axes, resulting in an internal/external rotation, upward/downward rotation, and anterior/posterior tilting (Figure 1). Commonly reported alterations in patients with injuries are increased anterior tilt, internal rotation, and decreased upward rotation.[18] These alterations are often the result of impaired muscle function: more anterior tilt as a result of increased tension in the Pm or decreased lower trapezius (LT) activity; increased internal rotation because of lower activity of the serratus anterior (SA) muscle; or decreased upward rotation as a result of decreased upper trapezius (UT) and SA activity, with increased tension in the LS.

Although a scapula-based program is considered to be relevant in the early phases of a rehabilitation program,[4] knowledge about scapular 3D kinematics combined with scapular muscle activity during exercises is lacking. Kibler et al[10] published a battery of 4 exercises useful during the early stages of rehabilitation. However, neither the activity of the deeper layer muscles nor the 3D scapular kinematics of those 4 exercises was investigated. In addition, the study did not measure the activity of the RC muscles.

Therefore, the purpose of this study was to extend the study of Kibler et al[10] and gain new insight into the muscle activity of the deeper layer scapulothoracic muscles.

METHODS

Participants

A total of 23 healthy male patients (mean age, 22.3 ± 1.5 years; mean height, 180 ± 0.06 cm; mean weight, 81.7 ± 17.2 kg; mean body mass index, 25.2 ± 5.5 kg/m^2) voluntarily participated in this study. The patients were included if the following criteria were met: patients were free of shoulder or neck problems in the past 6 months, had no fracture or surgery of the shoulder in the past, and did not perform overhead activities for more than 4 hours a week. All participants signed a written consent before testing. This study was approved by the ethical committee of the Ghent University Hospital (UZG 2016/1212).

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General Design

The 3D kinematics regarding the scapular rotations were measured and synchronized with EMG data of 7 scapulothoracic muscles (UT, middle trapezius [MT], LT, SA, LS, RM, Pm) and the musculus infraspinatus (IS) on the dominant side (side of writing hand) during the commonly used shoulder rehabilitation exercises as described by Kibler et al.[10] The exercises are described in Table 1. The patients randomly performed each exercise 5 times with 30 seconds of rest between the trials and 3 minutes of rest between the exercises. Two exercises were performed isometrically (inferior glide and low row), and 2 were dynamic (lawnmower and robbery). The isometric exercises were performed for a 5-second duration. The dynamic exercises were performed for 3 seconds during concentric movement, followed by a 5-second isometric hold. In view of comparative analysis of all 4 exercises, the 5-second isometric hold position at the end of the dynamic exercises was used for data analysis.

Test Procedure

A shoulder clinical evaluation was performed before preparing the patient for the measurements. This examination was composed of 5 tests to make sure the participants did not suffer from a shoulder disorder: Jobe, Neer, Hawkins, painful arc, or external rotation against resistance.[14] None of the patients reported any painful item during this examination.

The preparation for the electromyographic measurements consisted of shaving, scrubbing, and cleaning the patient's skin with alcohol to reduce impedance (<10 kΩ). A medical doctor specializing in fine-wire EMG (T. Parlevliet) placed the intramuscular paired hooked fine-wire electrodes (wire length, 125 mm; Chalgren) with ultrasound guidance into the following deeper layer muscles: LS, RM, IS, and Pm. A single-use 25-gauge hypodermic needle (length of the needle varying between 30 and 50 mm) was inserted in the muscle belly according to the location described by Delagi et al.[8] For the other 4 scapulothoracic muscles (UT, MT, LT, and SA), surface electrodes (Ambu Blue Sensor N – ECG electrodes; N-00-S/25) were placed with a 1-cm interelectrode distance according the instructions of Basmajian and De Luca.[1] Both surface and fine-wire electrodes were connected to a direct transmission system (DTS) EMG sensor (Noraxon USA Inc) with shielded cables. The cables were looped and taped on the skin to avoid movement artifacts or being accidently removed during the exercises. The EMG data were collected by a TeleMyo DTS Desk Receiver (Noraxon USA Inc) at a sampling rate of 3000 Hz.

A warm-up consisting of 15 repetitions of internal and external rotation in a neutral and 90° of abduction position and 15 repetitions of circumduction was followed by the performance of maximal voluntary isometric contractions (MVICs). The protocol of Castelein et al[7] was used for the scapulothoracic muscles (see Appendix Table A1, available in the online version of this article). For the MVIC of the IS, the patient was asked to perform an external rotation against resistance when sitting in an upright position

TABLE 1
Description: Positions of the Rehabilitation Exercises Including Pictures of the Exercises

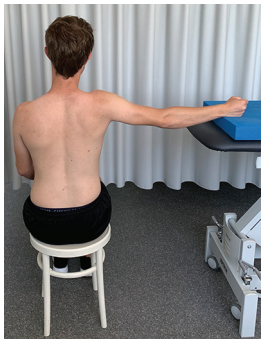
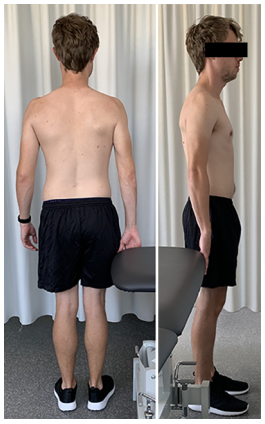
Exercise	Description
Inferior glide 	The patient sits with the arm 90° abducted and fist supported. The patient is asked to press with the fist in the direction of an arm adduction for 5 seconds.
Low row 	The patient stands in front of a bench with palm of the hand facing posteriorly. The patient is asked to push with the metacarpal part of the hand against the edge of the table for 5 seconds.
Lawnmower	The patient moves the arm into retraction and external rotation with the elbow in the direction of the back pocket while performing a trunk rotation and holds this end position for 5 seconds.
Robbery	The patient moves both arms into retraction and external rotation, with the elbow in the direction of the back pocket and holds this end position for 5 seconds.

TABLE 2
Placement of Reflective Spheres for 3D Scapular Kinematics[a]

Segment	Placement of the Marker
Head	LFHD-RFHD-LBHD-RBHD
Thorax	Processus spinosus C7 and T8–processus xiphoideus (most caudal point on the sternum) –processus coracoideus–deepest point IJ-LAC-RAC
Scapula	TS–AI–angulus acromialis (placed on cluster; most laterodorsal point of scapula)–scapula cluster (upper, front, and back cluster marker)
Arm	Upper arm cluster (PM-PL-DM-DL)–ME–LE–lower arm cluster marker (PM-PL-DM-DL)–processus styloideus ulna and radius
Pelvis	LSIAS-RSIAS-LSIPS-RSPIPS-LIC-RIC

[a]AI, angulus inferior; DL, distal-lateral; DM, distal-medial; IJ, incisura jugularis; LAC, left acromioclavicular joint; LBHD, left back head; LE, lateral epicondyle; LFHD, left front head; LIC, left iliac crest; LSIAS, left spina iliac anterior superior; LSIPS, left spina iliac posterior superior; ME, medial epicondyle; PL, proximal-lateral; PM, proximal-medial; RAC, right acromioclavicular joint; RBHD, right back head; RFHD, right front head; RIC, right iliac crest; RSIAS, right spina iliac anterior superior; RSIPS, right spina iliac posterior superior; TS, trigonum spinae.

with the arm against the body and elbow 90° flexed in the neutral position.[2] All positions were held for 5 seconds and repeated thrice.

After performing the MVIC, reflective spheres 12 mm in diameter were subsequently placed on bony landmarks at the dominant side, head, thorax, and pelvis of the patient for the measurement of the 3D kinematic data (Table 2). The kinematic data were collected by 6 Oqus 3+ and 2 Oqus 4 motion capture cameras supported by the Qualysis Track Manager (QTM; Qualysis AB) software. The marker capture frequency was set at 300 Hz. Both EMG and 3D kinematics were synchronously collected with QTM software during exercise performance.

Signal Processing and Data Analysis

Visual3D (Version 6 Professional, C-motion Inc) was used for the EMG and kinematic data processing of the different exercises. The kinematic patterns are described by the upper limb and pelvis model (Ghent University).[13,15,21,22] In this upper limb and pelvis model, the joint coordinate system for the shoulder girdle was based on the International Society of Biomechanics recommendation with a glenohumeral joint center estimation from scapular bony landmarks described by Meskers et al.[13] This model described the movements of the scapula around 3 different axes: horizontal axis (*X*-axis) resulting in anterior tilt or posterior tilt; frontal axis (*Y*-axis) resulting in upward rotation or downward rotation; and vertical axis (*Z*-axis) resulting in an external rotation or internal rotation of the scapula. All EMG signals were filtered with a high-pass Butterworth filter at 30 Hz and full-wave rectified. For the isometric exercises (inferior glide and low row) a DTS force sensor (Noraxon USA Inc), placed between the patient's hand and the bench, was used to determine when the patient started and ended pushing against the bench. For the isometric end position of the dynamic lawnmower and robbery, the windows of data were based on manually placed markers by the investigator at the start of the 5-second holding position and at the end of the movement. For all exercises, the first and last repetition of 5 were discarded. The average of the 5-second isometric hold of the second to fourth repetition was used for further data analysis with respect to the 3D scapular position and EMG activity for each muscle for all 4 exercises.

The MyoResearch 3.10 Master edition (Noraxon USA Inc) software was used for the signal processing of MVIC. Cardiac artifacts were subsequently removed from the signals, followed by a 30-Hz high-pass Butterworth filter, a full-wave rectification, and smoothing (root mean square of 100 milliseconds). For the MVIC analysis, a marker was manually placed at the start and after 3 seconds for each repetition. The mean amplitude of the 3 seconds between 2 markers was computed for each muscle at each position. For all muscles except the Pm and IS, the greatest mean EMG value over positions 1 to 4 (see Appendix Table A1, available online) was taken determining the MVIC used for the normalization of the EMG activity of the exercises. Concerning the IS and Pm, the EMG value of position 5 (see Appendix Table A1, available online) was used for the MVIC of Pm and the seated position with resistance against external rotation for the MVIC of IS.

Subsequently, all values for EMG activity were expressed as a percentage of MVIC (%MVIC). The values of EMG were categorized to facilitate the interpretation as low (<20% MVIC), moderate (20%-50% MVIC), or high (>50% MVIC) activity.[3]

Statistical Analysis

The statistical software SPSS version 25.0 (IBM Corp) was used for all statistical analysis. Means and standard deviations were calculated for the 3 scapular rotations (*X*-axis, *Y*-axis, and *Z*-axis) and normalized EMG activity for the 8 muscles investigated (UT, MT, LT, SA, LS, RM, Pm, and IS) during all 4 exercises. A linear mixed model for repeated measures with post hoc Bonferroni correction for pairwise comparison was used. Based on the residuals, data were normally distributed. The significance level was set at alpha = .05. Interaction effects exercise*scapular rotation and exercise*muscle were of primary interest. In case of absence of significant interaction effects, main effects for "scapular rotation," "muscle," and "exercise" were further explored.

RESULTS

Scapular Kinematics

Results of descriptive analysis are displayed in Table 3. A significant interaction effect for exercise*scapular rotation ($P \leq .001$) was found.

TABLE 3
Descriptive Statistics of the Scapular Positions During the Isometric Phase of the Rehabilitation Exercises[a]

	X-Axis (°) AT (–)/PT (+)	*Y*-Axis (°) DR (–)/UR (+)	*Z*-Axis (°) IR (–)/ER (+)
Inferior glide	–7.19 ± 6.96	23.91 ± 14.52	–31.30 ± 6.95
Low row	–8.83 ± 5.49	3.08 ± 5.13	–42.32 ± 15.04
Lawnmower	0.915 ± 4.89	29.24 ± 8.47	–14.97 ± 9.90
Robbery	–7.04 ± 5.79	30.32 ± 9.66	–13.34 ± 12.49

[a]AT, anterior tilt; DR, downward rotation; ER, external rotation; IR, internal rotation; PT, posterior tilt; UR, upward rotation.

The isometric phase of inferior glide, low row, and robbery resulted in a significantly ($P < .05$) increased anterior tilt position of the scapula compared with a slight posterior tilt position for the lawnmower. The low row displayed significantly ($P \leq .001$) less upward rotation compared with the other 3 exercises, where an upward rotation between 20° and 30° was reached. Besides the decreased upward rotation, the low row resulted in a significantly ($P \leq .001$) increased internal rotation relative to the other 3 exercises. In comparison with the lawnmower and robbery, the inferior glide as well as the low row represent significantly ($P \leq .001$) more internal rotation.

EMG Activity of Scapulothoracic Muscles

The results showed a significant interaction effect for position*EMG muscle activity ($P \leq .001$). The results of the pairwise comparison with the post hoc analysis are presented in Table 4. With respect to deep scapular muscles, LS and RM activity ranged between 27% and 66% MVIC. Pm activity was moderate (34%-44% MVIC) during low row and inferior glide compared with the lawnmower (9.53% MVIC) and robbery (11.85% MVIC).

DISCUSSION

The purpose of this study was to explore the differences between previously published exercises regarding electromyographic muscle activity and 3D kinematics of the scapula.[10] Combining scapular 3D kinematics measurements with fine-wire EMG measurements of the scapulothoracic muscle revealed new insights about the previously recommended exercises by Kibler et al[10] for early rehabilitation after shoulder injury.

TABLE 4
Detail of the Comparative Statistical Analysis for EMG Measurements Based on Exercises[a]

					P Value[b]	
Muscle	Inferior Glide	Low Row	Lawnmower	Robbery	Pairwise Comparisons	
UT	9.32 ± 5.16	7.39 ± 4.17	20.60 ± 12.23	23.49 ± 11.46	IG-LR: ≥.999	LR-LM: .287
					IG-LM: .521	LR-RB: .079
					IG-RB: .160	LM-RB: ≥.999
MT	18.38 ± 9.87	25.66 ± 12.39	36.21 ± 17.36	38.70 ± 22.18	IG-LR: ≥.999	LR-LM: .608
					IG-LM: .031	LR-RB: .251
					IG-RB: .009	LM-RB: ≥.999
LT	12.31 ± 5.83	13.26 ± 11.38	29.20 ± 14.90	24.42 ± 12.15	IG-LR: ≥.999	LR-LM: .075
					IG-LM: .048	LR-RB: .495
					IG-RB: .355	LM-RB: ≥.999
SA	20.27 ± 13.88	18.35 ± 17.92	20.42 ± 13.42	14.85 ± 11.93	IG-LR: ≥.999	LR-LM: ≥.999
					IG-LM: ≥.999	LR-RB: ≥.999
					IG-RB: ≥.999	LM-RB: ≥.999
LS	27.35 ± 22.62	36.37 ± 27.98	49.39 ± 34.89	58.05 ± 40.23	IG-LR: .917	LR-LM: .282
					IG-LM: .004	LR-RB: .004
					IG-RB: <.001	LM-RB: .952
RM	27.51 ± 14.10	27.02 ± 14.95	58.68 ± 25.17	66.12 ± 39.66	IG-LR: ≥.999	LR-LM: <.001
					IG-LM: <.001	LR-RB: <.001
					IG-RB: <.001	LM-RB: ≥.999
Pm	44.36 ± 37.65	34.06 ± 38.43	9.53 ± 10.99	11.85 ± 22.20	IG-LR: .618	LR-LM: <.001
					IG-LM: <.001	LR-RB: .002
					IG-RB: <.001	LM-RB: ≥.999
IS	23.79 ± 15.21	12.73 ± 8.18	36.58 ± 30.67	30.13 ± 21.23	IG-LR: .617	LR-LM: ≤.001
					IG-LM: .279	LR-RB: .036
					IG-RB: ≥.999	LM-RB: ≥.999

[a]Values are expressed in %MVIC ± SD. EMG, electromyography; IG, inferior glide; IS, infraspinatus; LM, lawnmower; LR, low row; LS, levator scapulae; LT, lower trapezius; MT, middle trapezius; MVIC, maximal voluntary isometric contraction; Pm, pectoralis minor; RB, robbery; RM, rhomboid major; SA, serratus anterior; UT, upper trapezius.
[b]$P < .05$ indicates statistical significance.

Comparing our results with the Kibler et al[10] study, comparable activity levels were found for the UT, LT, and SA, illustrating the similarity between studies regarding common outcome variables. Interestingly, new insights were noted in the results of the activity in the deeper layer muscles (Pm, RM and LS, and the IS) and the 3D kinematics of the scapula.

The MT (36%-39%) and RM (59%-66%) showed their greatest activity at the isometric phase of the dynamic exercises. The greater activity of the RM compared with the MT can be clarified by the moderate upward rotation position during the exercises. Possibly, MT would be more active in higher degrees of upward rotation of the scapula.[11] The LS and RM revealed high activity of 50% to 66% during the isometric phase of the dynamic exercises. We asked patients to retract their scapula during this phase, which explains why the RM shows this amount of activity. The LS seems to help the RM in this function. The muscle activity of the Pm was the greatest (34%-44%) during the inferior glide and low row, both isometric exercises against a fixed resistance. Similar results were reported by Castelein et al[6] during a series of closed-kinetic chain exercises with fixed hands.

In the protective phase of nonoperative or postoperative rehabilitation programs, less activity of the IS is recommended.[17] Therefore, the inferior glide, lawnmower, and robbery exercises should be avoided because of the moderate activity of the IS. The low row had the least activity (13%) for the IS, making the exercise suitable in the early phase of rehabilitation of shoulder disorders.

All 4 exercises except the lawnmower showed an anterior tilt of the scapula. For the inferior glide and low row, the position of the scapula can be explained by the higher amount of EMG activity of the Pm. However, for the robbery this position cannot be explained by increased Pm activity, but possibly rather by the instruction to bring the elbows *down* during this exercise. The upward rotation is related to the position of the arm; namely, greater upward rotation values are described in exercises with a relative arm elevation. Therefore, the low row showed the least degree of upward rotation. The lawnmower and robbery revealed lesser degrees of internal rotation of the scapula (15°) compared with the inferior glide and low row. Those results can be elucidated by the greater activity of the RM, as this muscle is expected to participate in this movement.

Strengths and Limitations

The major strength of this study lies in the complex methods used to analyze muscular activity simultaneously with 3D scapular kinematic behavior during commonly used exercises. The combination of fine wire with surface electrodes synchronously measured with the 3D kinematics successfully provides us new information about scapular behavior during the exercises. However, some limitations should be noted.

First, only healthy patients were included in this study. Therefore, some caution should be taken when generalizing the results. We can also assume that some discomfort could have appeared during the performance of a movement when fine wires are used. However, while performing the exercises none of the patients were limited in arm motions because of discomfort.

Second, although the procedure was performed as standardized as possible, some inconsistencies may have influenced our results. The same medical doctor (T. Parlevliet) performed all fine-wire applications, and the same researcher (K.B.) performed the resistance during the MVIC for all participants. Furthermore, all participants had to perform the exercises with the same time cues to standardize the movement as much as possible (see Appendix Table A2, available online). Notwithstanding, the performance of the exercises could not be totally standardized. To minimize the differences, each patient was placed in the same position at the beginning of the exercise as well as informed in the same way regarding how he was supposed to execute the movement.

Future studies should include other exercises to gain more information about synchronized measured 3D scapular position and EMG activity. As we only measured the IS, because of technical limitations, the other RC muscles of the shoulder girdle were not explored in this study. As the muscles could be reached with fine-wire EMG, researchers should consider exploring the other RC muscles in the future as well.[1,8]

Clinical Implications

Based on the results of this study, several clinical implications can be drawn to guide the clinician in the exercise choice for scapular rehabilitation.

(1) In patients who exhibit insufficient posterior tilt and/or hyperactivity of the Pm, all exercises, except the lawnmower, are contraindicated based on increased anterior tilt of the scapula and/or increased activity in Pm. Only the lawnmower leads to a posterior tilt of the scapula and reveals EMG activity of only 9.5% in the Pm.
(2) For patients with insufficient upward rotation, and possibly hyperactivity in LS, the exercises are not recommended, given the moderate to great activity of LS for all 4 exercises.
(3) For patients with increased internal rotation, the lawnmower and robbery may be advised since these exercises place the scapula in a more external position compared with the common resting position in the scapular plane (approximately 30° anterior of the frontal plane).[18] The other 2 exercises are contraindicated.

In summary, the lawnmower contains scapular positions in the direction of posterior tilt, upward rotation, and external rotation combined with less Pm activity, compared with the other exercises. On the other hand, the low row resulted in an anterior tilt scapular position with less upward rotation and great internal rotation in combination with a moderate activity (±35%) of the LS and Pm. Three of the exercises (inferior glide, lawnmower, and robbery) should not be included in the early phase of

rehabilitation because of the moderate activity of the IS. However, the clinician should take into account that this study only focused on local scapular muscle activity, whereas in the clinical setting the patient should always be managed from a biopsychosocial context.

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